

1. PRG $G: \{0,1\}^n \rightarrow \{0,1\}^{l(n)}$

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Alice constructs:

CPTS 327 HW 3

$$G^{**}: (\{0,1\}^n)^m \rightarrow (\{0,1\}^{l(n)})^m \text{ for } x_1, x_2, \dots, x_m \text{ where } x_i \in \{0,1\}^n$$
$$G^{**}(x_1, \dots, x_m) := G(x_1) \parallel G(x_2) \parallel \dots \parallel G(x_m) \text{ // concatenates all output}$$

Is G^{**} a secure PRG?

↳ can an adversary distinguish from random?

- G^{**} is composed of outputs of PRG G . If G outputs strings indistinguishable from random, G^{**} is indist. from rand (PRG).

↳ A

2. PRG $G: \{0,1\}^1 \rightarrow \{0,1\}^{l(1)}$, fixed polynomial $l(1) > 1$

Bob's $G^n: \{0,1\}^1 \rightarrow \{0,1\}^{l(n)}$, flips every bit
 $G^n(s) := G(s) \oplus 1^{l(n)}$ } is secure PRG?

↳ Yes, by reduction. If G is PRG (original msg. cannot be derived by the adversary), $\parallel$ distinguished from rand, flipping the bits of the ciphertext will reveal no new info about the original msg.

↳ A

3. PRG $G: \{0,1\}^n \rightarrow \{0,1\}^{2n}$ is secure. // independent string

Carol's $G^*: \{0,1\}^n \rightarrow \{0,1\}^{2n}$, input seed: $s \in \{0,1\}^n, r \in \{0,1\}^n$

$$G^*(s) := G(s \oplus r)$$

↳ No, by C

The XOR w/random r violates determ.

4. PRG $G: \{0,1\}^{2n} \rightarrow \{0,1\}^{l(n)}$ is secure w $l(n) > 2n$

David's $G': \{0,1\}^{2n} \rightarrow \{0,1\}^{l(n)}$, $s \in \{0,1\}^{2n}$, $s = s_1, \dots, s_{2n}$ ($s_i \in \{0,1\}$)
 $G'(s) := G(s_1, \dots, s_n \parallel 0^n)$

No, by D.

The mapping from seed space $\rightarrow$ cipher text doesn't involve an XOR or masking operation, only the first n bits of a $2n$ msg. are encrypted

5. $F: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$ is a secure PRF

$F': \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$ as:

$$F'(k, x) = F(k, x) \oplus F(k, x \oplus 1^n)$$

World 0 ($b=0$): The challenger samples uni. key $k \leftarrow \{0,1\}^n$ and computes $y_i = F'(k, x_i)$

World 1 ($b=1$): The challenger samples truly rand. $F \leftarrow \text{Funs}[\{0,1\}^n, \{0,1\}^n]$
and computes $y_i = F(x_i)$

a) $p_1 = p_r [b'=0 \mid b=1]$ (Alice outputs 0 in world 1)

↳ Alice finds i, j st. $y_i = y_j$ (for $i \neq j$), from truly random

$$\text{Total Queries} = \binom{10}{2} = 45$$

$$\text{Prob. } i \neq j = (1 - 2^{-10})^{45} = 1 - \frac{45}{1024} \approx 0.956$$

$$p_1 = 1 - 0.956 = 0.0439$$

b) $p_0 = P_r[b'=0 | b=0]$ (Alice outputs 0 in world 0), from PRF F'

↳ every input and its complement collide w/prob. 1, 5 complementary pairs $\therefore p_0 = 1$

c) $\text{Adv} = [P_1 - P_0] = [1 - 0.0439] = \frac{979}{1024} = 0.956$

d) No, because $F'(x) = F'(\bar{x})$ (symmetry of F)

↳ F' is not a secure PRF

6. $G: \{0,1\}^n \rightarrow \{0,1\}^{n+1}$ be a secure PRG.

PRG $G': \{0,1\}^n \rightarrow \{0,1\}^{\ell}$ as follows:

- $s_0 \leftarrow \{0,1\}^n$

- $j = 1, \dots, \ell$: compute $s_j b_j \leftarrow G(s_{j-1})$, where b_j is last bit

- output string: $y = b_1 b_2 \dots b_{\ell}$

Assume PPT adversary A distinguishes $G'(s)$ from a truly random string with advantage ϵ . $\ell := 5$

GAME 0 v. GAME 1:

- Game 0 is the correct starting point

Game 1: b_1 and s_0 are random

$s_1 b_1' = G(s_0)$

↳ $G(s_0)$ computed in game 0 as well

- b_1 is replaced by a truly random bit

↳ 1 is valid, from PRG security, $G(s_0) = U_{n+1}$, replacing b_1 with random bit is indistinguishable from the original b_1' . (both game output strings appear rand.?)
- only one change made.

GAME 2: $b_1, b_2 \leftarrow \{0,1\}^2$ and $s_2 \leftarrow \{0,1\}^n$ // each random

for $j=3 \dots 5$, $s_j b_j = G(s_{j-1})$

$(s_2, b_2) = G(s_1)$ // from game 1

$G(s_1) \rightarrow U_{n+1}$

// Game 2 replaces $s_2 b_2$ PRG call w/ uniform random, which is allowed by PRG security

$\therefore$ valid

GAME 3:

- b_3 is also random

- $s_3 b_3' = G(s_2)$ follows the conventions on game 1

All changes (b_3 rand) are permitted by PRG security. $\therefore$ valid.

GAME 4: $b_1, b_2, b_3, b_4 \leftarrow \{0,1\}^4$ // b_4 is also random

$s_4 \leftarrow \{0,1\}^n$ // random

$s_5 b_5 = G(s_4)$, which corresponds to replacing $(s_4, b_4) = G(s_3)$ with uniform. (s_4) is uniform.

All additions follow conventions of the previous games, $\therefore$ valid.

GAME 5: b_1, b_2, b_3, b_4, b_5 are fully random
(replaces $(s_3, b_5) = G(s_4)$ w/random

↳ s_4 is already uniform from game 4. $\therefore$ game 5 is valid.

7. Game 0 is a correct starting point. Identify each game's correctness
(From Hybrid 0/real world to Hybrid $i+1$ /Hybrid n random function)

$G: \{0,1\}^l \rightarrow \{0,1\}^{2l}$ // length doubling PRG

$G(s) = G_0(s) \parallel G_1(s)$ // output parsed into first/last l bits

Hybrid 0: root key k is random

- for each query x , compute full path using PRG G each level

Hybrid i ($1 \leq i \leq n$)

↳ Instead of deriving nodes via PRG, nodes at depth i are random

($R_{x_i} \leftarrow \{0,1\}^l$) and nodes at depth $i+1$ are computed with $G(R_{x_i})$

For Hybrid $i+1$, method of generating nodes at depth $i+1$ are ALSO assigned randomly as well

↳ Hybrid 0 to i and i to $i+1$ are valid they replace 1 more level of the tree from PRG generation to actual random gen.

Hybrid $n-1 \rightarrow n$ // nodes at $n-1$ are random and nodes at depth n are generated w/PRG G (following hybrid i)

Hybrid n has fully random node generation (drawing from hybrid $i+1$)

↳ Because we replace only one PRG call with fully random assignment each call, all games from H_0 to H_n are valid and computationally indistinguishable.